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0178/01

October/November 2014

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS **MUST NOT** BE USED IN THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 80.

This document consists of **16** printed pages.



1 Evaluate each of the following.

(a) $7 - 2 \times 8 + 1$

Answer (a) [1]

(b) 0.42×0.03

Answer (b) [1]

2 (a) Write the following numbers in order of size, starting with the **smallest**.

$\frac{2}{7}$

28.5%

0.281

$\frac{7}{25}$

Answer (a) < < < [1]

- (b) A bag of corn has mass 2.7 kg.
650 grams of corn is taken out of the bag.

Calculate the mass of the corn which remains in the bag.
Give your answer in kg.

Answer (b) kg [1]

3 (a) On a certain day, the temperature at midnight was recorded as -9°C and the temperature at noon was recorded as 17°C .

Find the difference between the two temperatures.

Answer (a) $^{\circ}\text{C}$ [1]

- (b) A farmer erects a 48 m fence along one side of his yard.
He places poles at 3 m intervals along the fence.

How many poles will he need?

Answer (b) [1]

4 Evaluate each of the following.

(a) $\frac{2}{3} + \frac{1}{7}$

Answer (a) [1]

(b) $3\frac{1}{2} \div 4$

Answer (b) [1]

5 $a = -2$, $b = 3$ and $c = 1$.

Find the value of the following.

(a) $\frac{4b - 3a}{3c}$

Answer (a) [1]

(b) $a^2 - ac + b$

Answer (b) [1]

6 The table below shows the timetable for a bus from Maseru to Butha-Buthe.

	Maseru	Teyateyaneng	Hlotse	Butha-Buthe
Arrive	—	0644	0801	0915
Depart	0600	0713	0832	—

(a) How long does the bus stop in Teyateyaneng?

Answer (a) mins [1]

(b) What is the total time spent **driving** from Maseru to Butha-Buthe?
Give your answer in hours and minutes.

Answer (b) hrs mins [1]



- 7 (a) Solve the inequality $4 - 2x < 1$.

Answer (a) [1]

- (b) State the smallest possible integer that satisfies $4 - 2x < 1$.

Answer (b) [1]

- 8 The number of spherical marbles, N , that can be made from a lump of clay varies inversely as the cube of the diameter, d mm, of the marbles made.
When the diameter of the marbles is 2 mm, 64 marbles can be made from the lump of clay.

Find

- (a) the formula for N in terms of d ,

Answer (a) $N =$ [2]

- (b) the number of marbles with diameter 4 mm that can be made from the lump of clay.

Answer (b) [1]

- 9 Solve the simultaneous equations.

$$5x + 2y = 14$$

$$2x - 3y = 17$$

Answer $x =$

$y =$ [3]

10 $\frac{1}{r} = \frac{2}{s} + \frac{3}{t}$

Express s in terms of r and t .

Answer $s =$ [3]



- 11 (a) The table shows the number of people who attended a stadium to watch a football match on three consecutive weekends.

Weekend	Number of people who attended
1	5.08×10^4
2	7.13×10^3
3	1.01×10^5

- (i) Work out the total number of people who watched the matches over the three weekends.

Answer (a)(i) [1]

- (ii) A ticket for a match costs M200.00.

Work out the amount of money collected from ticket sales on the second weekend.
Give your answer in standard form.

Answer (a)(ii) M [1]

- (b) $p = 4 \times 10^{-4}$ and $q = 3 \times 10^3$.

Calculate pq .

Answer (b) [1]

- 12 The base of a building is a rectangle with external dimensions of length 40 m and width 30 m, measured correct to the nearest metre.

(a) Write down the upper bound of the length of the building.

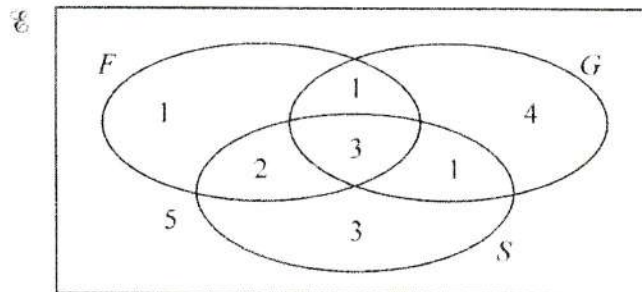
Answer (a) m [1]

(b) The thickness of each wall of the building is 1 m, measured correct to the nearest 10 cm.

Calculate the maximum internal length of the building.

Answer (b) m [2]

13



In the Venn diagram,

$E = \{\text{pupils in the class}\}$

$G = \{\text{pupils who are girls}\}$

$S = \{\text{pupils who are swimmers}\}$

$F = \{\text{pupils born on a Friday}\}$

(a) Find $n(F \cap G \cap S)$.

Answer (a) [1]

(b) A pupil is chosen at random from the class.

Find the probability that the pupil

(i) is a swimmer,

Answer (b)(i) [1]

(ii) is a boy swimmer, born on a Friday.

Answer (b)(ii) [1]



14 (a) Simplify $(5x^2)^3$.

Answer (a) [1]

(b) Evaluate each of the following.

(i) $5^3 + 5^2 + 5^1 + 5^0$

Answer (b)(i) [1]

(ii) $16^{\frac{3}{2}}$

Answer (b)(ii) [1]

15 A straight line passes through the points $A(5, 6)$ and $B(3, -2)$.

Find

(a) the co-ordinates of the midpoint of AB ,

Answer (a) (.....)(1)

(b) the gradient of AB ,

Answer (b) [1]

(c) the equation of AB .

Answer (c) [2]

- 16 (a) Six positive integers have a mean of 5, a median of 6 and a range of 7.

Write down the six integers in numerical order.

Answer (a) [2]

- (b) Express as a single fraction.
Give your answer in its simplest form.

$$\frac{3}{x-1} - \frac{x}{x+3}$$

Answer (b) [2]

17 $A = \begin{pmatrix} 2 & -1 \\ 4 & -1 \end{pmatrix}$ $B = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$

Find

(a) $B - 2A$,

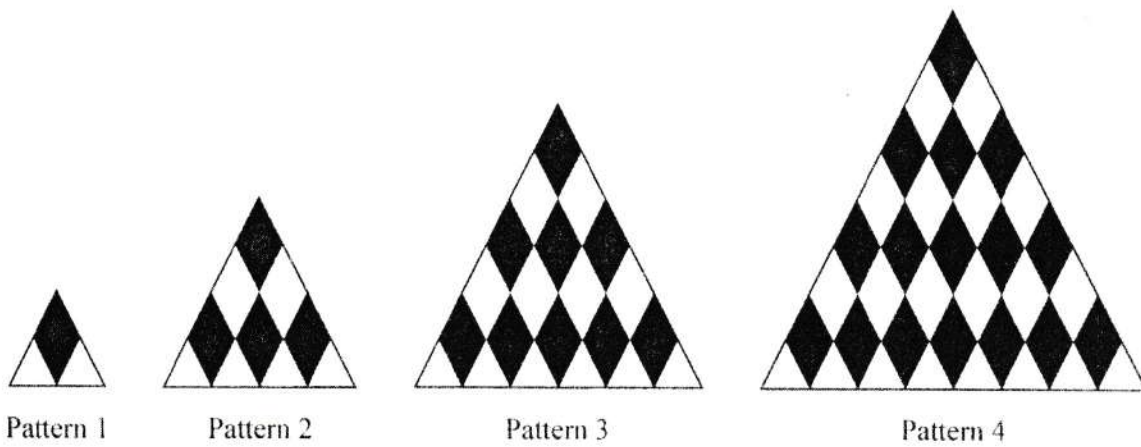
Answer (a) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(b) A^{-1} .

Answer (b) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]



18 The diagram shows patterns of triangles and rhombuses.



(a) Complete the table for Pattern number 5.

[1]

Pattern number	Number of unshaded triangles	Number of unshaded rhombuses	Number of shaded rhombuses	Total number of rhombuses
n	r	u	s	t
1	2	0	1	1
2	4	2	4	6
3	6	6	9	15
4	8	12	16	28
5				

(b) Find an expression for

(i) s in terms of n ,

Answer (b)(i) $s = \dots\dots\dots$ [1]

(ii) t in terms of n .

Answer (b)(ii) $t = \dots\dots\dots$ [2]

19 $f(x) = 4 + x^2$ $g(x) = \frac{2x-3}{5}$

Find the following.

(a) $f(3)$

Answer (a) [1]

(b) $g^{-1}(x)$

Answer (b) [2]

(c) $gf(x)$

Give your answer in its simplest form.

Answer (c) [2]



- 20 (a) Construct triangle PQR with $PQ = 7$ cm, $QR = 6$ cm and angle $PQR = 70^\circ$.
The line PQ has been drawn for you.



[2]

- (b) Construct, using a ruler and a compass only, the locus of points inside triangle PQR which are

(i) equidistant from PQ and QR ,

[1]

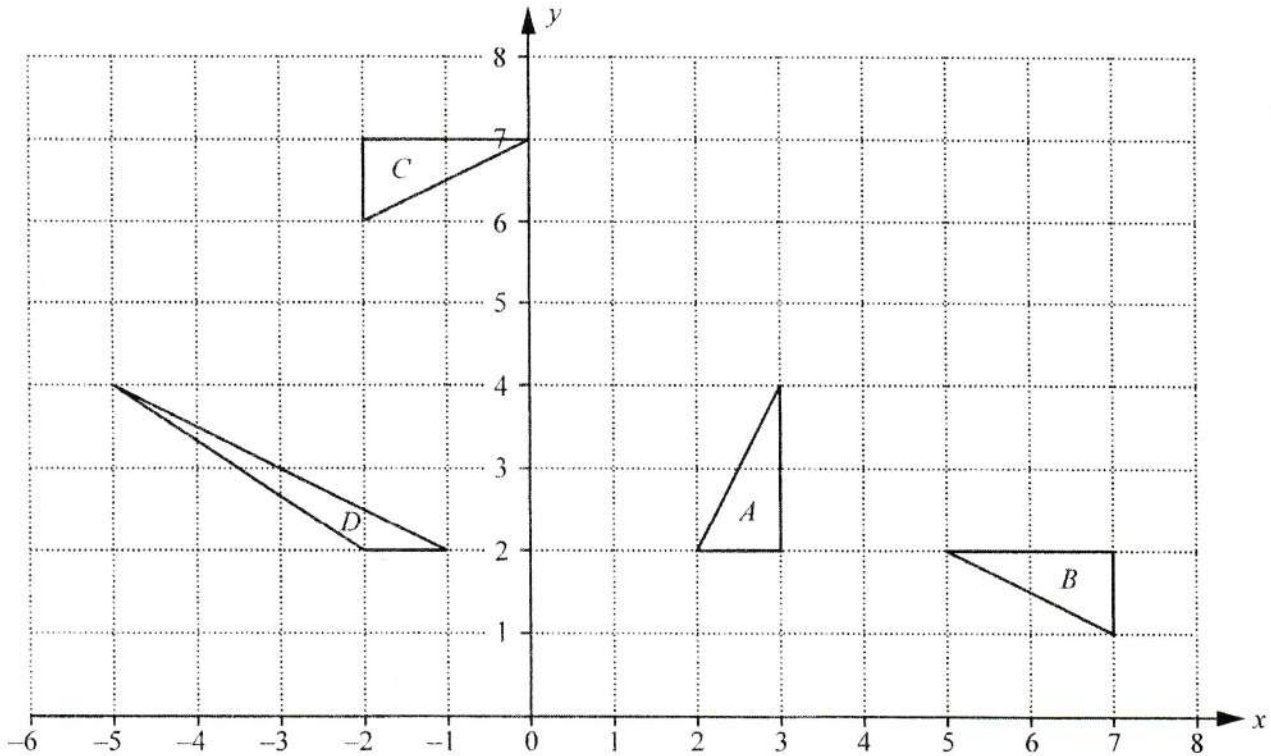
(ii) equidistant from P and Q .

[1]

- (c) Shade the region, inside triangle PQR , containing the points that are nearer to PQ than QR and nearer to Q than P .

[1]

- 21 The diagram shows triangles A , B , C and D .



- (a) Triangle A is mapped onto triangle B by a rotation.

Write down the centre of rotation.

Answer (a) (.....,)[1]

- (b) Describe fully the **single** transformation that maps triangle A onto triangle C .

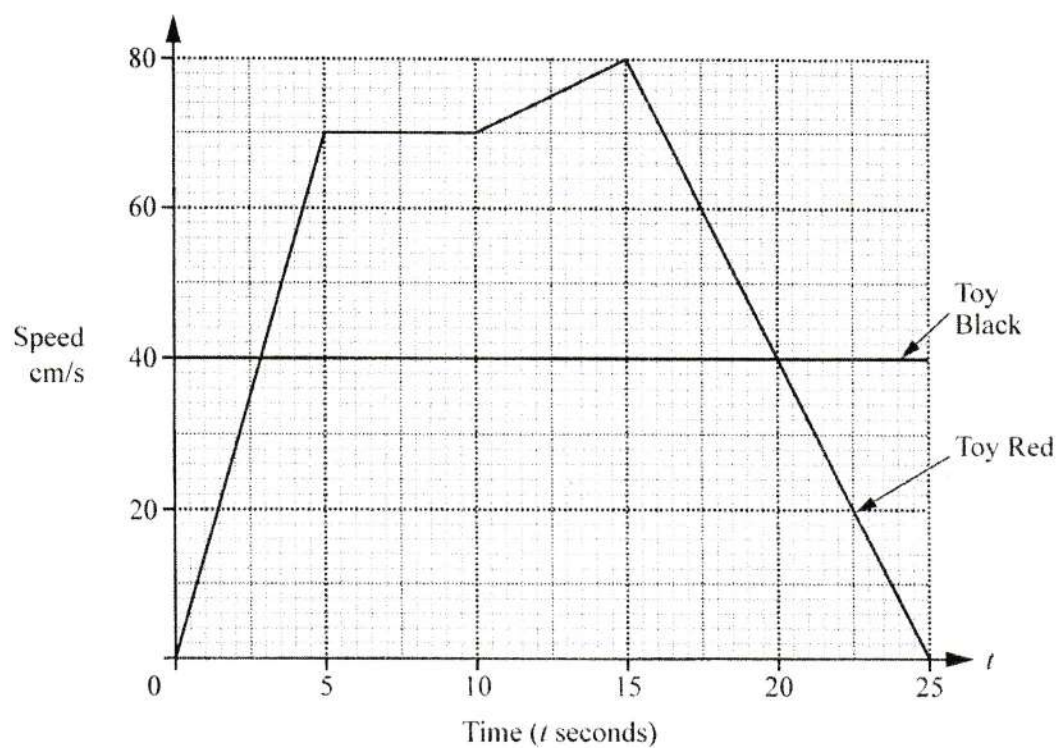
Answer (b)
[2]

- (c) Write down the matrix which represents the transformation that maps triangle A onto triangle D .

Answer (c) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]



- 22 The diagram shows the speed-time graphs of two remote controlled toy cars, Toy Red and Toy Black.



- (a) Describe the motion of Toy Red between $t=10$ and $t=15$.

Answer (a) [1]

- (b) Find the times at which both cars travel at the same speed.

Answer (b) secs [1]

- (c) (i) Calculate the area under the graph for Toy Red between $t = 0$ and $t = 10$.

Answer (c)(i)..... [2]

- (ii) What does this area represent?

Answer (c)(ii)

..... [1]

- (d) Calculate the time, between $t = 5$ and $t = 10$, when both cars had travelled the same distance.
Give your answer as a mixed number in its simplest form.

Answer (d)..... secs [3]

Question 23 is printed on the next page.



- 23 At a market Fako buys maize at M45 per bag and peas at M75 per bag. One day Fako had M900 to spend.

(a) Taking m for the number of maize bags bought and p for the number of pea bags bought, write an inequality and show that it reduces to $3m + 5p \leq 60$.

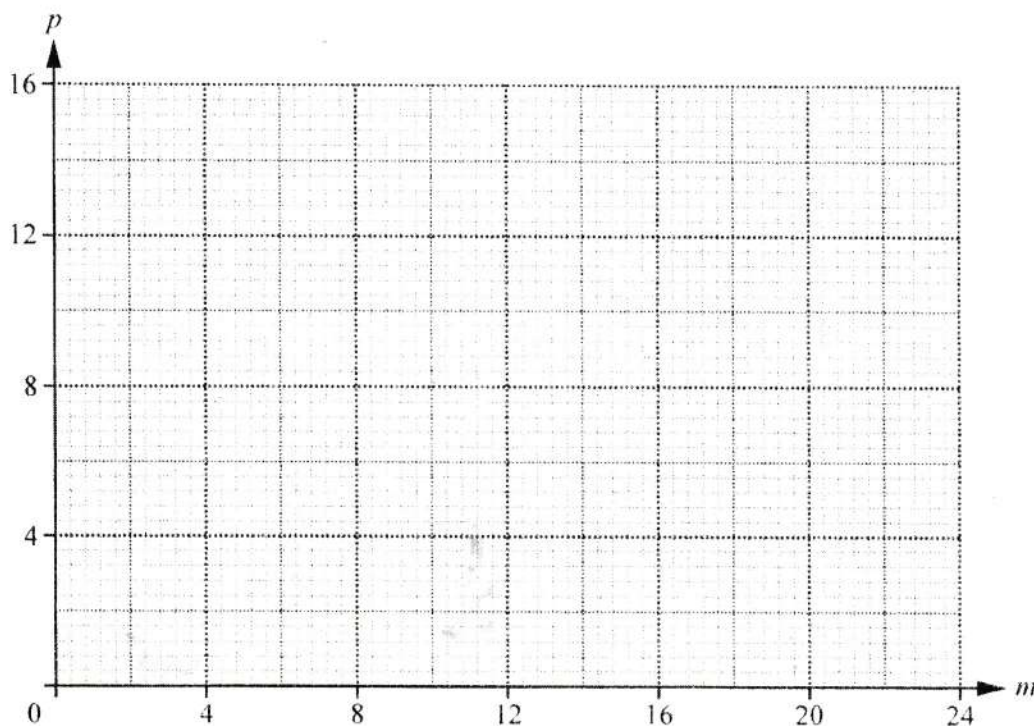
[1]

(b) Fako decided to buy at least 4 bags of peas so $p \geq 4$. He also decided to buy at least 10 bags in total.

Write the third inequality satisfied by m and p .

Answer (b) [1]

(c) Represent these three inequalities on the grid below and shade the unwanted region.



[3]

(d) Use the graph to find the **maximum total** number of bags that Fako can buy.

Answer (d) [1]